

Scenario-Based Report

Prompt Engineering Techniques

A Comparative Study Using ChatGPT, Gemini & Claude

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1. Introduction

Prompt Engineering is the practice of designing and refining input instructions — called **prompts** — to elicit accurate, relevant, and high-quality responses from Large Language Models (LLMs) such as GPT, Claude, and Gemini. A prompt is the gateway between human intent and machine output. The quality, structure, and framing of a prompt can dramatically alter the usefulness of the AI response.

This report examines **four foundational prompt engineering techniques**: Straightforward Prompts, Tabular Format Prompting, Missing Word Prompting, and Preceding Question Prompting. Each technique is explained with its definition, core principles, practical examples, use cases, advantages, limitations, and an illustrative diagram.

2. Straightforward Prompts

2.1 Definition & Overview

Straightforward Prompting is the most fundamental technique. It involves providing a **clear, direct, single-sentence or single-paragraph instruction** without special formatting or contextual scaffolding. The core principle is simplicity: *ask clearly, receive clearly*.

2.2 Core Principles

- **Clarity:** The prompt must be unambiguous and grammatically correct
- **Specificity:** Include enough context to constrain the response appropriately
- **Conciseness:** Avoid unnecessary padding or repetition in the prompt
- **Action Orientation:** Begin with an action verb: Explain, Translate, Summarize, List, Define

2.3 Flow Diagram



Figure 1: Straightforward Prompting – End-to-End Flow from Query to Response

2.4 Practical Examples

Translation Task

Prompt: "Translate the following English sentence into Spanish: 'Artificial Intelligence is transforming the world.'"

Response: "La Inteligencia Artificial está transformando el mundo."

Mathematical Task

Prompt: "Calculate the area of a circle with radius 7 cm."

Response: "Area = $\pi \times r^2 = 3.14159 \times 49 \approx 153.94 \text{ cm}^2$ "

2.5 Advantages & Limitations

Aspect	Advantages	Limitations
Complexity	Low cognitive load	Too vague for complex tasks
Speed	Fastest prompt type to write	No iterative refinement
Output	Clean, focused responses	May lack structure for multi-part answers
Applicability	Works across all AI models	Not for nuanced analysis tasks

3. Tabular Format Prompting

3.1 Definition & Overview

Tabular Format Prompting is a technique where the user explicitly instructs the AI to present its response as a **structured table** with clearly defined rows and columns. This approach is ideal for comparisons, reference data, and multi-attribute descriptions, leveraging the brain's ability to process tabular data significantly faster than dense paragraphs.

3.2 Core Principles

- **Explicit Format Instruction:** Always specify 'in a table' or 'as a table with columns'
- **Column Specification:** Define the exact columns you need for maximum relevance
- **Row Scope:** Specify how many items or what category of items to include
- **Sortability Intent:** Optionally specify ordering (e.g., sorted by year, alphabetically)

3.3 Structure Diagram

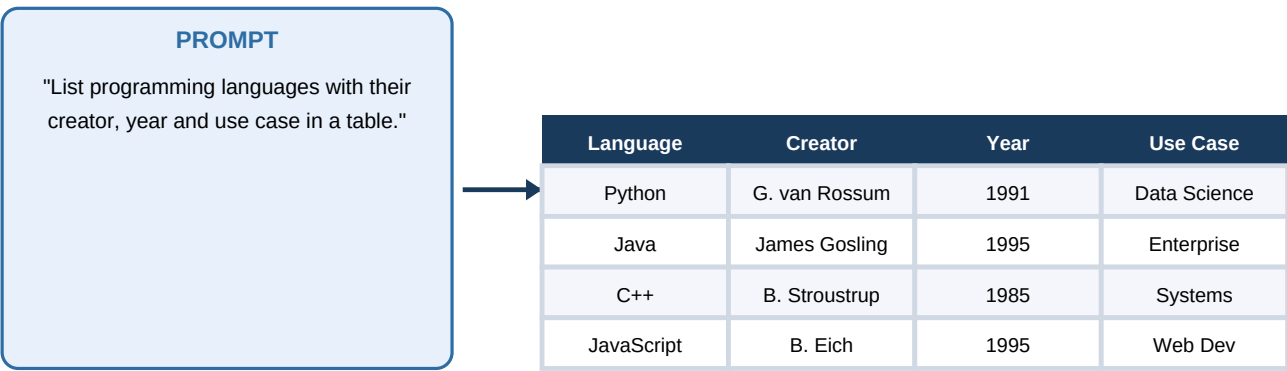


Figure 2: Tabular Format Prompting – From Instruction to Structured Table Output

3.4 Example – Programming Languages

Prompt: "Create a table comparing 5 popular programming languages with columns: Language, Creator, Year, Primary Use Case, and Difficulty Level."

Language	Creator	Year	Primary Use	Difficulty
Python	G. van Rossum	1991	Data Science / AI	Beginner
Java	James Gosling	1995	Enterprise Apps	Intermediate
C++	B. Stroustrup	1985	Systems Programming	Advanced
JavaScript	Brendan Eich	1995	Web Development	Beginner
Swift	Apple Inc.	2014	iOS Development	Intermediate

3.5 Example – AI Tools Comparison

AI Tool	Developer	Launch Year	Key Strength	Best Use Case
ChatGPT	OpenAI	2022	Broad versatility	General writing & coding
Claude	Anthropic	2023	Safety & reasoning	Analysis & summarization
Gemini	Google	2023	Multimodal ability	Search & image tasks

4. Missing Word Prompting

4.1 Definition & Overview

Missing Word Prompting (also called **Fill-in-the-Blank** or **Cloze Prompting**) provides an incomplete sentence or template with blanks, asking the AI to fill in missing content based on context and world knowledge. This technique ensures the response integrates naturally into existing text by constraining AI output to a specific syntactic and semantic structure.

4.2 Core Principles

- **Context Anchoring:** Surrounding text provides strong constraints on the answer
- **Grammatical Fit:** The filled word must match grammatical role (noun, verb, adjective)
- **Semantic Coherence:** The answer must make logical sense within the full sentence
- **Knowledge Retrieval:** The model draws from its training data to supply accurate facts

4.3 Mechanism Diagram

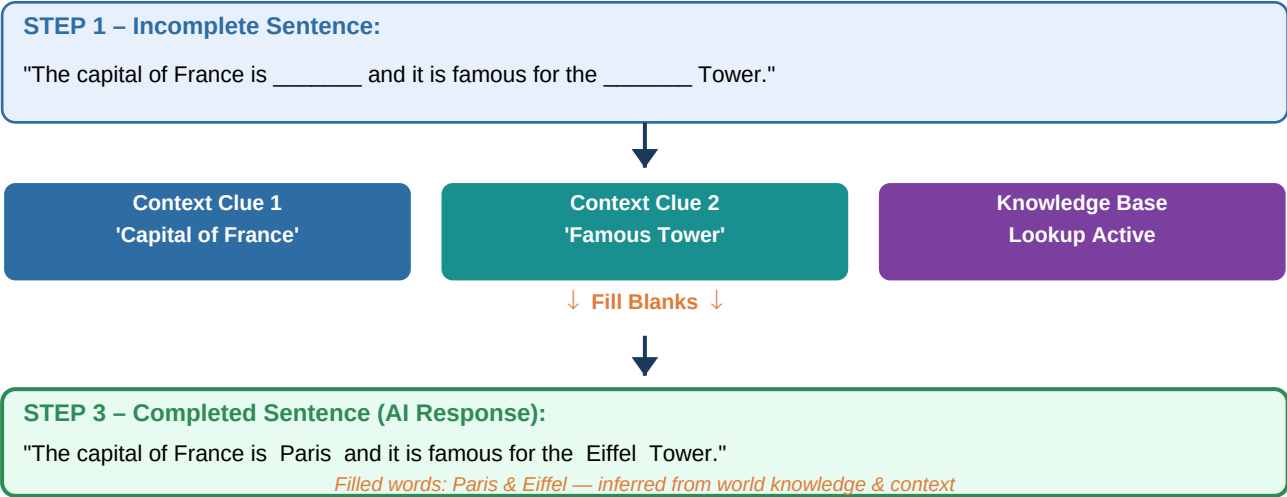


Figure 3: Missing Word Prompting – Three-Step Fill-in-the-Blank Mechanism

4.4 Types of Missing Word Prompts

Type	Prompt Example	AI Response
A – Single Blank (Factual)	Speed of light is approx. _____ km/s	299,792 km/s
B – Multiple Blanks	Capital of Japan is _____ and currency is the _____	Tokyo ... Yen
C – Creative	There lived a _____ who could speak to _____	wizard ... dragons
D – Technical	my_list = _____	[1, 2, 3, 4, 5]
E – Scientific	Force = _____ × _____	mass × acceleration

5. Preceding Question Prompting

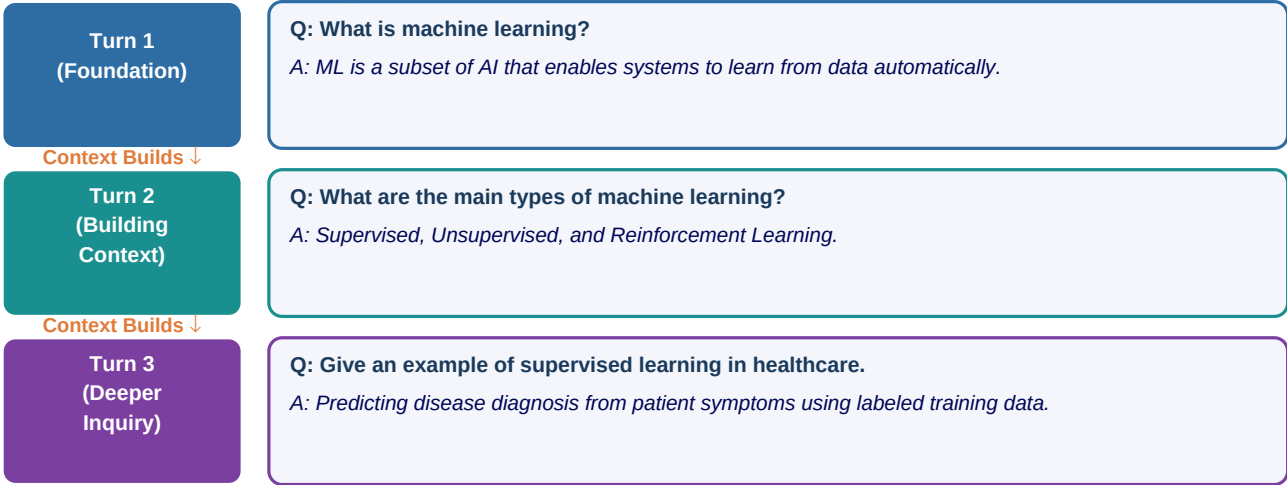
5.1 Definition & Overview

Preceding Question Prompting (also known as **Sequential Questioning** or **Chain Prompting**) asks a series of related questions in a deliberate sequence, where each subsequent question builds on the context established by the previous question and answer. This mirrors the Socratic method of inquiry, progressively uncovering deeper knowledge through targeted questions.

5.2 Core Principles

- **Sequential Logic:** Questions follow a logical progression from broad to specific
- **Context Retention:** Each response informs and constrains the next question
- **Progressive Depth:** Earlier questions establish foundations; later ones build expertise
- **Coherent Narrative:** The entire sequence maintains a unified topic thread

5.3 Context Flow Diagram



Each subsequent question leverages the context of all preceding Q&A pairs for more precise answers.

Figure 4: Preceding Question Prompting – Multi-Turn Context Accumulation Flow

5.4 Example – Blockchain Technology

Turn	Question (Prompt)	Purpose
1	What is blockchain technology?	Foundation — establish core concept
2	How does a block get added to the chain?	Mechanism — understand the process
3	Difference between public and private blockchain?	Classification — build taxonomy
4	How does blockchain ensure data immutability?	Deep dive — explore security property
5	Real-world example of blockchain in supply chain?	Application — practical relevance

6. Comparative Analysis

The following table compares all four techniques across seven critical dimensions, enabling practitioners to select the most appropriate technique for any given task.

Dimension	Straightforward	Tabular Format	Missing Word	Preceding Question
Complexity	Low	Low–Medium	Medium	High
Output Type	Free-form text	Structured grid	Filled template	Evolving narrative
Best For	Quick facts	Comparisons	Templates	Deep research
Context Use	Minimal	Minimal	Moderate	Extensive
User Effort	Very Low	Low	Medium	High
Response Quality	Good	Very Good	Very Good	Excellent
Learning Curve	None	Minimal	Low	Moderate

Selection Guidance

- **Straightforward Prompts:** Use for immediate, factual, single-answer queries where speed matters
- **Tabular Format:** Use when you need to compare multiple items across consistent attributes
- **Missing Word Prompting:** Use when you have a template or structure and need AI to populate it
- **Preceding Question:** Use when conducting deep research, learning, or complex analysis

Technique Complexity & Effort Visual Comparison



Figure 5: Relative complexity level of each technique

Structural Patterns in Preceding Question Prompting

Pattern Name	Description	Example Domain
Funnel Pattern	Start broad, progressively narrow	Research & Analysis
Pyramid Pattern	Build from facts to synthesis	Academic Studies
Spiral Pattern	Return to topics with deeper framing	Philosophy & Ethics
Diagnostic Pattern	Isolate the problem through elimination	Technical Debugging
Comparative Pattern	Alternate between two items	Decision Making

Best Use Cases – All Techniques

Straightforward	Tabular Format	Missing Word	Preceding Question
Quick factual lookups (definitions, formulas)	Comparing products or technologies	Educational assessments & quiz generation	Complex multi-faceted research topics
Simple translation tasks	Displaying reference data (capitals, elements)	Code completion & programming assistance	Learning new technical subjects step by step
Basic calculations & unit conversions	Summarizing survey results or findings	Legal & business document template filling	Debugging complex software progressively
Summarizing short passages	Creating schedules and timelines	Language learning & grammar correction	AI-assisted interviews or discovery sessions

7. Conclusion

This report has provided a comprehensive examination of four foundational prompt engineering techniques. Each technique occupies a distinct niche in the AI interaction toolkit, and the most effective prompt engineers are those who can fluidly move between them based on task requirements.

Straightforward Prompts	Remain the workhorse for everyday AI interactions, valued for their speed and simplicity.
Tabular Format Prompting	Transforms AI into a powerful data organization engine, ideal for comparison and reference tasks.
Missing Word Prompting	Bridges AI capability with human template design, creating a powerful hybrid output mode.
Preceding Question Prompting	Unlocks the deepest level of AI collaboration, enabling sophisticated research through structured dialogue.

As Large Language Models continue to evolve, prompt engineering will remain a critical bridge between human cognitive intent and machine intelligence. Future research directions include combining these techniques, studying their relative effectiveness across LLM architectures, and developing standardized prompt libraries for domain-specific professional applications.

8. References

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